

ASSESSMENT – 1

CompTIA A+

*Hardware Components
& System Assembly*

Section A — Concept Application

SCENARIO 1: You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

Question: Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.



Selecting Compatible Hardware for an Office Workstation

1. **Motherboard:** Select a motherboard that supports the required processor socket, RAM generation, and storage connections.
2. **CPU:** Choose a processor that is compatible with the motherboard's socket and chipset.
3. **RAM:** Verify the RAM type (DDR4 or DDR5), supported speed, memory capacity, and number of slots available on the motherboard.

4. Storage: Ensure that the motherboard has the required storage connectors, such as M.2 NVMe or SATA, for the selected SSD or HDD.
5. Compatibility Check: Refer to the manufacturer's specifications to verify processor support, RAM compatibility, expansion slots, and power supply requirements before purchasing the components.

Conclusion: For a workstation to function properly, all hardware components must be compatible in terms of sockets, interfaces, supported standards, and power requirements.

SCENARIO 2: You are advising a client whose employees mainly run office and productivity applications on their workstations.

Question: Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

➤ Comparison of HDD, SSD, and NVMe Storage Technologies

Storage Type	Technology	Speed	Cost	Best Use
HDD	Uses rotating magnetic disks to store data.	Slow	Low	Storing large amounts of data at a low cost.
SSD	Uses flash memory and commonly connects through SATA.	Fast	Moderate	Everyday office tasks and general computing.
NVMe SSD	Uses flash memory with a PCIe interface.	Very fast	Higher	Highperformance tasks and faster data access.

Recommendation: For office computers used for document preparation, spreadsheets, presentations,

and other productivity tasks, an NVMe SSD is a suitable choice.

Technical Reasons:

- Provides significantly faster data transfer speeds than traditional HDDs and SATA SSDs.
- Offers low latency, allowing applications and files to load quickly.
- Uses the PCIe interface, which supports higher data bandwidth.
- Reduces system startup time and improves multitasking performance.
- Contains no moving mechanical parts, resulting in silent operation and better resistance to physical shock than HDDs.

Conclusion: An NVMe SSD with a capacity of 500 GB to 1 TB is a practical storage option for modern office workstations, offering a good balance of speed, responsiveness, and storage capacity.

SCENARIO 3: You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.



Explanation:

- GPU (Graphics Processing Unit): The GPU is responsible for processing visual data and graphics-related operations. A good-quality GPU helps design applications run smoothly, speeds up rendering, and improves performance when working with 3D models and complex graphics.
- Cooling System: The cooling system helps maintain a safe temperature by dissipating the heat generated by the CPU and GPU during extended usage. Proper cooling avoids overheating, reduces thermal throttling, prevents unexpected system shutdowns, and protects hardware components from damage.

Conclusion: A powerful GPU combined with an efficient cooling system helps maintain smooth operation, system stability, and reliable performance even during demanding tasks performed over long periods.

SCENARIO 4: You are assembling a PC from components sourced from three different vendors.
Question: Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

➤ **Importance of Hardware Compatibility in PC Assembly**

Hardware compatibility plays a vital role in building a computer because every component must be able to work properly with the other parts to ensure smooth and reliable operation.

Why Hardware Compatibility Is Important:

- Confirms that the processor fits the motherboard socket and is supported by its chipset.
- Ensures that the RAM's generation, capacity, and speed are compatible with the motherboard.
- Verifies that the power supply unit (PSU) can deliver sufficient power to all components.

- Checks whether storage devices can be connected through the available SATA or M.2 interfaces.
- Makes sure that the graphics card is compatible with the motherboard and fits inside the computer case.
- Helps avoid system errors, performance issues, and unexpected failures.

Problems Caused by Incompatible Components:

1. The computer may fail to start or complete the boot process.
2. Installed RAM might not be recognized by the system.
3. An unsupported processor may prevent the motherboard from functioning correctly.
4. Improper cooling or power delivery can lead to overheating.
5. An underpowered PSU may cause sudden shutdowns, crashes, or instability.
6. Components may not fit properly within the PC cabinet due to size limitations.
7. Certain hardware features may become unavailable because of incompatible drivers or connection standards.

Conclusion: Before purchasing computer components from different manufacturers, it is necessary to verify their socket types, supported technologies, physical dimensions, power requirements, and motherboard compatibility. This helps build a stable, efficient, and dependable computer system.

SCENARIO 5: You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

Question: Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

➤ **PSU Selection and Safety Precautions**

1. Selecting the Appropriate PSU

- Determine the total power consumption of the CPU, GPU, motherboard, storage drives, and other connected devices.

- Select a power supply that provides approximately 20–30% additional wattage to handle peak loads and future upgrades.
- Verify that the PSU includes all the necessary power connectors for the installed hardware.
- Choose a good-quality PSU with built-in protection against overvoltage, overcurrent, short circuits, and overheating.

2. Safety Precautions

- Always turn off and disconnect the PSU from the electrical supply before installing or removing any connections.
- Inspect all cables and hardware components for physical damage before use.
- Follow anti-static precautions to protect sensitive electronic components from electrostatic discharge.
- Use only cables specifically designed for the selected PSU to avoid damaging the hardware.

3. Connecting the PSU

- Attach the 24-pin ATX connector to the motherboard's main power socket.

- Connect the 8-pin CPU power cable to the appropriate motherboard connector.
- Supply power to the graphics card using the required PCIe or other supported power connectors.
- Connect SATA power cables to the SSDs and HDDs.
- Make sure every connector is firmly fitted and properly secured.

4. Checks Before the First Boot

- Verify that the motherboard and all other components are correctly installed.
- Remove any loose screws or objects that could cause a short circuit.
- Ensure that power cables are properly arranged and kept away from cooling fans.
- Recheck all power connections to confirm that they are fully inserted.
- Finally, connect the AC power cable and turn on the PSU to prepare the system for startup.

Conclusion: Selecting a PSU with adequate wattage, suitable connectors, and essential safety features is necessary for reliable PC operation. Proper installation and thorough safety checks before switching on the

system help prevent electrical faults and hardware damage.

SCENARIO 6: You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

➤ **POST (Power-On Self-Test) Checks**

1. Steps to Verify POST:

1. Turn on the computer and observe whether the cooling fans start running and the indicator lights turn on.
2. Check whether the monitor displays the BIOS or POST screen during startup.
3. Ensure that no error messages or abnormal beep sequences appear.

4. Open the BIOS setup and verify that the processor, RAM, storage drives, and graphics card are properly recognized.

5. Confirm that the system starts successfully and loads the operating system without any issues.

2. Troubleshooting When POST Fails:

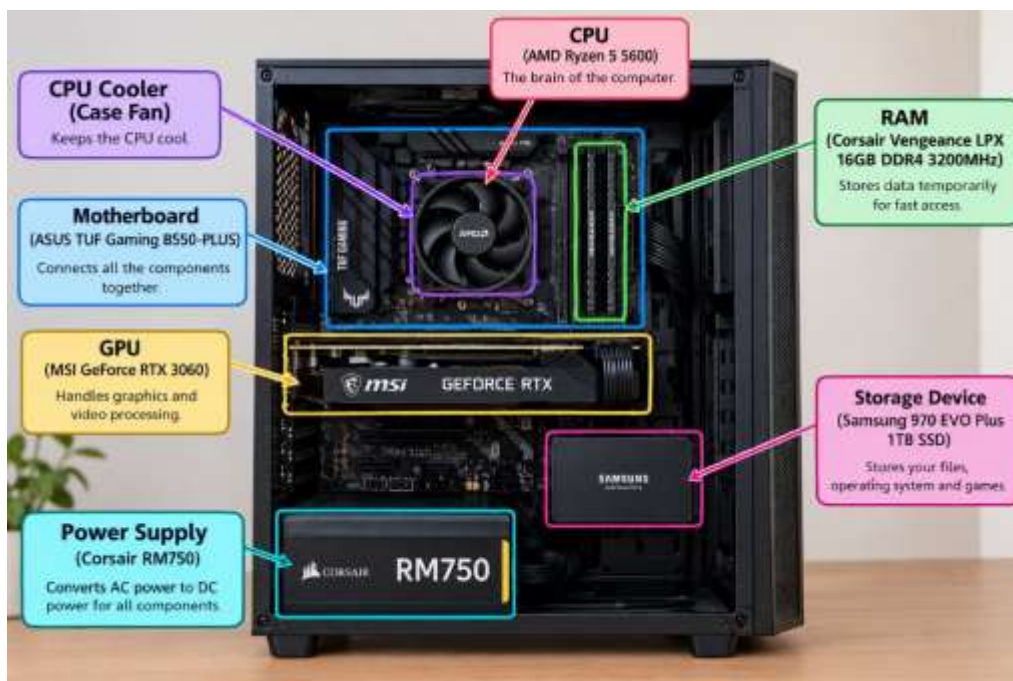
- Shut down the computer and disconnect it from the power source before inspecting the hardware.
- Examine all power cables and component connections to ensure they are properly attached.
- Remove and reinstall the RAM modules and graphics card to establish proper contact.
- Observe the motherboard's diagnostic LEDs or beep codes to identify possible hardware faults.
- Reset the BIOS settings or clear the CMOS if necessary.
- Identify the defective component through testing and replace it if required.

Conclusion: A workstation can be considered ready for client use once it successfully completes POST, recognizes all essential hardware components, and starts the operating system without displaying errors.

Section B — Practical Coding Tasks

Task 1: Component Identification & Functions

- Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system.
- Note the make/model or key specification of each component where visible.
- Present the findings as a labeled component list or table.



Desktop Computer Components and Their Functions

No.	Component	Function	Make/Model or Key Specification
1	Motherboard	Acts as the main circuit board, connecting all hardware components and enabling communication between them.	MSI B450 TOMAHAWK
2	CPU (Central Processing Unit)	Executes instructions, performs calculations, and manages the operations of the computer.	AMD Ryzen

3	RAM (Random Access Memory)	Temporarily holds data and programs that are currently being accessed by the processor.	Corsair RAM
4	GPU (Graphics Processing Unit)	Handles graphical operations, including image processing, video playback, and rendering.	MSI GeForce RTX
5	Storage (SSD)	Stores the operating system, applications, and user data for long-term use.	Samsung 970 EVO Plus
6	Cooling System	Dissipates heat generated by the processor and maintains a safe operating temperature.	AMD CPU Cooler

Conclusion: These six components perform different but interconnected functions, including processing, temporary memory, graphics rendering, data storage, hardware communication, and temperature control. Together, they ensure the desktop computer operates efficiently and reliably.

Task 2: Motherboard Ports & Socket Identification

- Identify and label all major motherboard ports (SATA, power, USB headers, etc.).
- Identify all expansion slots (PCIe, M.2, etc.) and their purpose.
- Identify the CPU socket type and note which CPU families are compatible with it.

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Slot	Purpose
PCIe x16	Mainly used for installing a graphics card and other high-bandwidth expansion devices.
PCIe x1	Used for smaller expansion cards such as WiFi, sound, and capture cards.

M.2 Slot	Supports high-speed NVMe SSDs and, on some motherboards, other M.2 devices such as Wi-Fi modules.
DIMM Slots	Used to install RAM modules on the motherboard.

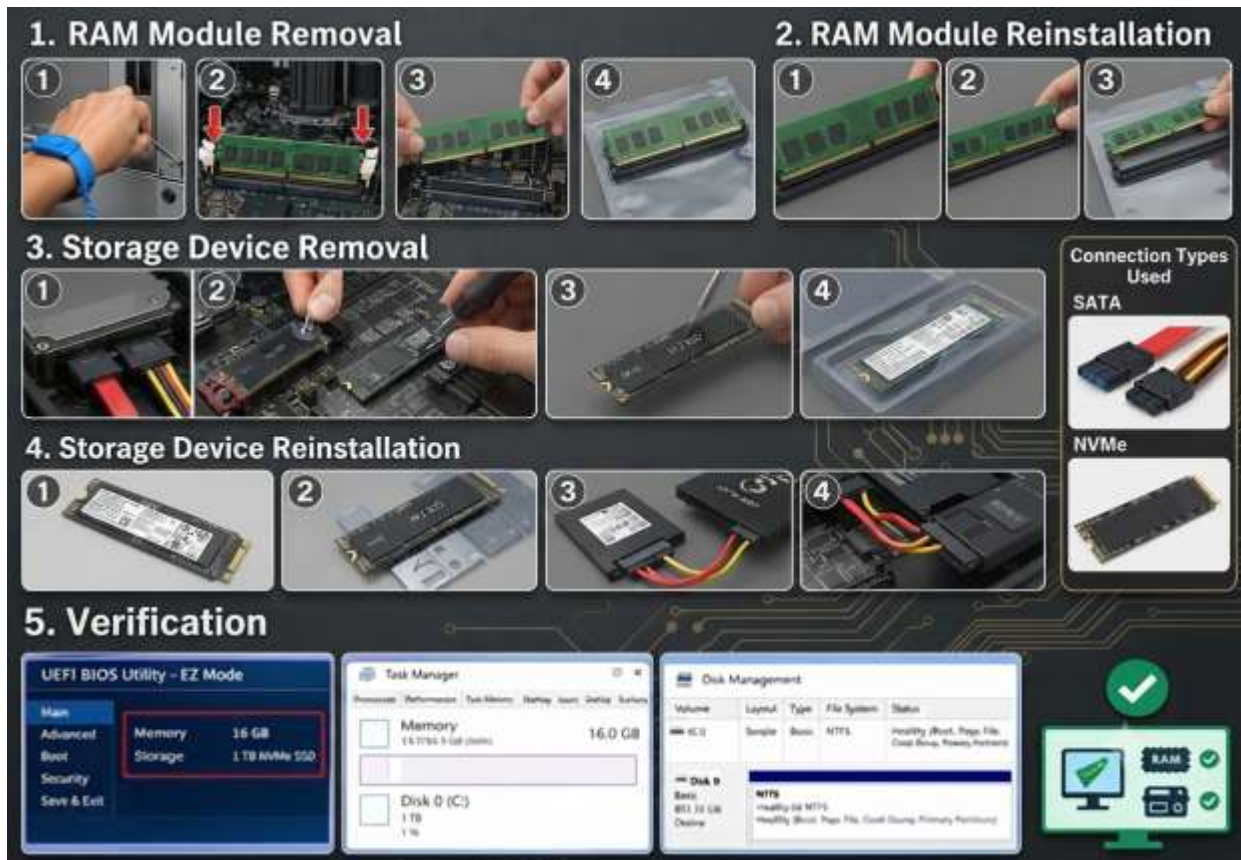
CPU Socket Compatibility

- **Socket Type:** AM4
- **Compatible Processors:** AMD Ryzen processors from the 1000 to 5000 series can be supported, depending on the particular motherboard chipset and BIOS version.
- Before installing a processor, the motherboard's CPU support list and BIOS version should be checked.

Conclusion: The motherboard provides the necessary sockets and slots for the CPU, RAM, graphics card, storage, and other expansion devices. The exact compatibility and available ports can vary according to the motherboard model and its specifications.

Task 3: RAM & Storage Installation

- Safely remove the RAM module, noting ESD precautions taken.
- Reinstall the RAM module correctly and confirm it is seated properly.
- Remove and reinstall the storage device, noting the connection type used (SATA/NVMe).
- Confirm both components are detected correctly after reinstallation.



RAM & Storage Installation

1. Removing and Reinstalling RAM

- First, shut down the computer completely and remove the power cable.
- Use an anti-static wrist strap or touch a grounded metal surface to avoid electrostatic discharge (ESD).
- Open the clips on both sides of the RAM slot and gently take out the RAM module.
- Match the notch on the RAM module with the corresponding notch in the motherboard slot.

- **Place the RAM into the slot and press it down evenly until the side clips lock automatically.**
- **Finally, make sure the RAM is properly seated and both clips are secured.**

2. Removing and Reinstalling Storage

- **Turn off the computer and disconnect it from the power supply.**
- **For a SATA SSD or HDD, remove the SATA data and power cables, take out the drive, and reconnect the cables properly after reinstalling it.**
- **For an NVMe SSD, remove its mounting screw and carefully take out the drive. Insert the SSD into the M.2 slot at a slight angle and then secure it with the screw.**
- **Check that all connections are properly fitted before starting the computer.**

3. Checking Whether the Components Are Detected

- **Start the computer and open the BIOS/UEFI settings.**
- **Check whether the correct amount of installed RAM is shown.**
- **Verify that the SSD, HDD, or NVMe drive is listed under the storage or boot-device section.**

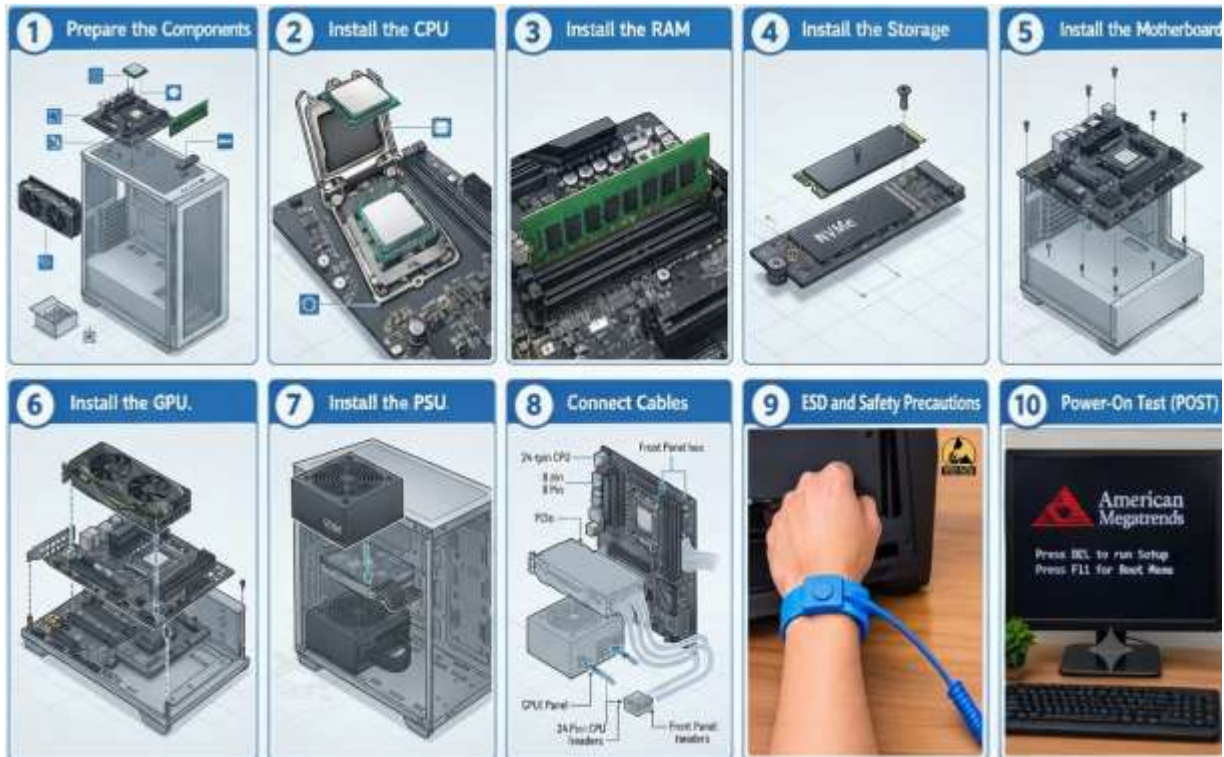
- **After entering the operating system, check whether the storage device can be accessed normally.**

Conclusion:

The RAM and storage devices can be safely removed and installed again by following proper handling and ESD precautions. Correct installation can be confirmed when the RAM and storage drive are recognized properly in the BIOS/UEFI and operating system.

Task 4: Full Desktop PC Assembly

- Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU.
- Follow correct installation sequence and ESD/safety precautions throughout.
- Document each step with labeled photos or diagrams.
- Power on the system and confirm it passes POST successfully.



Full Desktop PC Assembly Procedure

1. First, gather all the required components and tools needed for assembling the computer.
2. Carefully place the CPU into the correct socket on the motherboard.
3. Install the RAM modules into the appropriate memory slots and make sure they are locked properly.
4. Mount the required storage device, such as an SSD or HDD.

5. Place the motherboard inside the cabinet and secure it using the appropriate screws.
6. Insert the graphics card into the suitable PCIe slot and fix it properly.
7. Install the power supply unit (PSU) in its designated position inside the cabinet.
8. Connect the motherboard, CPU, GPU, storage, and other required components using the correct power and data cables.
9. Take proper ESD precautions throughout the assembly process to prevent damage to sensitive components.
10. Connect the power cable, switch on the computer, and verify that the system completes the POST process without errors.

Conclusion:

The desktop computer was assembled by installing and connecting all the major hardware components correctly. After completing the assembly, the system was powered on and successfully passed the POST check.

Section C — Mini Capstone Project

Mini Project: Workstation Deployment & Boot Failure Diagnosis

1. identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.



Three Important Things to Check if a PC Does Not Boot

If the computer does not start, I would first inspect these three areas:

1. **Power Supply (PSU):** Check whether the PSU is switched on and supplying power to the system.
2. **RAM:** Make sure the RAM modules are properly inserted and locked in their slots.
3. **CPU and Motherboard:** Check that the processor is correctly installed and supported by the motherboard.

2. Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.

➤ **Checking CPU and Motherboard Compatibility**

The processor and motherboard must be compatible with each other. The following points should be verified:

- Check whether the CPU socket matches the motherboard socket.
- Confirm that the motherboard supports the particular CPU family.
- Check whether the required BIOS version supports the processor.

If the processor is not supported by the motherboard, the system may fail to start or complete the boot process.

3. Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.



Checking RAM and Storage Devices RAM:

- Verify that the RAM is correctly fitted into the motherboard slot.
- Open the BIOS/UEFI and check whether the installed memory capacity is detected correctly.

Storage:

- For SATA drives, check both the SATA data and power connections.
- For an M.2 SSD, make sure it is properly inserted and secured in the M.2 slot.
- Check the BIOS/UEFI to confirm that the storage device is recognized.

4. Produce a hardware-installation checklist of at least three steps to prevent similar failures in future deployments.



PC Installation Checklist

Before switching on the computer, the following points should be checked:

- Follow proper ESD protection measures.
- Confirm CPU and motherboard compatibility.
- Ensure that RAM is installed correctly.
- Verify proper installation of the storage device.
- Connect all necessary PSU cables.
- Recheck the connections and cables.
- Turn on the PC and check the POST/BIOS screen.

5. Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.



Advantages of SSD for Office Computers

An SSD is a suitable choice for office computers because it provides faster performance than a traditional HDD.

- The system boots more quickly.
- Programs and applications open faster.
- Files can be accessed in less time.
- SSDs have no mechanical moving parts.
- They operate quietly.
- They generally provide better responsiveness during everyday tasks.

Conclusion:

An SSD, particularly an NVMe SSD, can provide fast and responsive performance for modern office computers and is well suited for regular productivity work.

Section D — AI-Augmented Learning

STEP 1 · BUILD WITH AI

- Guides a junior technician through diagnosing a desktop PC that fails to boot immediately after assembly, in the correct priority order.
- Covers at minimum the motherboard, RAM, storage, and PSU as possible failure points.
- Is presented as a clear checklist or simple decision flow a technician could follow on site.
- Is applied to the ABC Solutions scenario from Section C to show which step would have caught the reported issue.



Prompt Given to AI AI

Prompt:

Prepare a simple troubleshooting checklist for a junior technician to identify the reason why a newly assembled desktop PC is not starting. Include safety precautions, PSU, motherboard, CPU, RAM, storage,

and POST checks. Arrange the steps in a logical order and use easy-to-understand language.

Checklist Generated by AI

The AI suggested the following basic steps:

1. Check whether the PC is getting power.
2. Inspect the PSU and its cables.
3. Examine the motherboard.
4. Check whether the CPU is installed correctly.
5. Inspect the RAM.
6. Check the graphics card.
7. Verify the storage device.
8. Start the PC and observe the POST screen.
9. Look for error codes or other warning indications.

My Corrected Checklist

After checking the AI's response, I made some changes and added a few important steps:

1. Switch off the PC, unplug it from the power supply, and take proper ESD precautions.

2. Check the PSU and make sure all power connections are properly attached.
3. Inspect the motherboard and CPU installation.
4. Verify that the CPU and motherboard are compatible.
5. Remove and reseat the RAM properly.
6. Check the GPU and make sure its power cable is connected if required.
7. Inspect the storage device and its connection.
8. Start the computer and check the POST/BIOS screen.
9. Observe the motherboard's diagnostic LEDs or listen for beep codes.
10. If the problem is still present, test the components individually to find the possible faulty part.

What I Changed

I placed **ESD safety** at the beginning because handling PC components without proper precautions can cause damage. I also added a separate check for **CPU and**

motherboard compatibility, which was not clearly included in the AI-generated checklist. I rearranged the steps so that safety and hardware connections are checked before moving to POST troubleshooting. These changes make the checklist more practical and suitable for a junior technician.